National University of Computer and Emerging Sciences



Laboratory Manuals

for

Computer Networks - Lab

(CL -3001)

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Lab Manual 02

Objective:

- To learn basic commands of Linux related to Directory and File Manipulation, Process management and Network Management
- Introduction to Wireshark
- Analyzing the HTTP packets using Wireshark

Instructions:

- Submit one PDF/Word document containing answers and relevant screenshots.
- File name: `RollNo_Lab02`
- Submit through LMS before the announced deadline.
- Screenshots must clearly show the relevant Wireshark packet/filter or terminal output.

Part 1: Linux Commands

- **Find out the purpose of the following commands and execute them on your system with different parameters. [5]**

  ls, cd, pwd, mkdir, rmdir, rm , cp, mv, touch

- **Some commands may be new for most of you. Practice these terms on your own: [9]**

top	The top program provides a dynamic real-time view of a running system. It can display system summary information as well as a list of tasks currently being managed by the Linux kernel.	top
ps	ps displays status of a selection of the active/currently running processes.	ps
kill pid	The **kill** command is used to send a signal to a process, where **PID** stands for Process ID. The general syntax is **`kill [-signal] PID`**. To view the list of available signals, use the command **`kill -l`** in the terminal. For example, **`kill`**	kill –SIGKILL pid

	`-SIGTERM 1234` sends a termination signal to the process with PID 1234. The `SIGKILL` signal can be used to forcefully terminate a process when necessary, but it should be used carefully and only for processes you are authorized to terminate.	
chmod	This command is used to grant or revert reading, writing, and executing permissions from a user, group or others. Following are the symbolic representation of three different roles: You can check the details by typing **man chmod** chmod 400 lab1.txt Check what happened to your file. Now write chmod 700 lab1.txt What happened to your file?	
ifconfig	ifconfig is used to configure the kernel-resident network interfaces. If no arguments are given, ifconfig displays the status of the currently active interfaces. If a single interface argument is given, it displays the status of the given interface only; if a single -a argument is given, it displays the status of all interfaces, even those that are down. Otherwise, it configures an interface.	ifconfig ifconfig -a ifconfig eth0
route	Route manipulates the kernel's IP routing tables. Its primary use is to set up static routes to specific hosts or networks via an interface.	route
ss	The command is used to investigate socket statistics. Use ss -u for udp and ss -t for tcp sockets to analyze which sockets are being used for which protocol.	
wget	wget stands for "web get". It is a command-line utility which downloads files over a network. It supports HTTP, HTTPS, and FTP protocols, as well as retrieval through HTTP proxies. wget	

	has been designed for robustness over slow or unstable network connections; if a download fails due to a network problem, it will keep retrying until the whole file has been retrieved. If the server supports rejects permission, it will instruct the server to continue the download from where it left off. The simplest way to use wget is to simply provide it with the location of a file to download over HTTP. For example, to download the file http://website.com/files/file.zip, this command: wget http://website.com/files/file.zip Where will this file be downloaded?	
traceroute	**Traceroute** is a network diagnostic tool used to determine the path taken by packets from the source host to a destination. It sends probe packets with increasing TTL values and records the intermediate routers that respond. Write on your terminal traceroute nu.edu.pk	
nslookup	nslookup is a network administration tool for querying the Domain Name System (DNS) to obtain domain name or IP address mapping or any other specific DNS record. It is also used to troubleshoot DNS related problems. Write on your terminal window nslookup www.google.com	
host	It is an alternative of nslookup but with more details. Write up on your terminal window: host www.google.com	

- **Ping command: [5]**

  Use the ping command to test connectivity to at least five reachable hosts/IP addresses, including the loopback address 127.0.0.1. Record the average round-trip time (RTT) reported by ping and compare the results.

Syntax for Pinging is:

ping [other network's ID (Domain/IP Address)]

Question - You are required to ping at least 5 other networks (including your own address i.e. 127.0.0.1) and compare the latency rate of all networks.

Part 2: Wireshark

Introduction to Wireshark

Wireshark, a network analysis tool formerly known as *Ethereal*, captures packets in real time and display them in human-readable format. Wireshark includes filters, color-coding and other features that let you dig deep into network traffic and inspect individual packets.

- **Running Wireshark**

When you run the Wireshark program and start capturing the internet traffic onto a specific interface (e.g., Ethernet, WiFi), the Wireshark graphical user interface will be displayed.

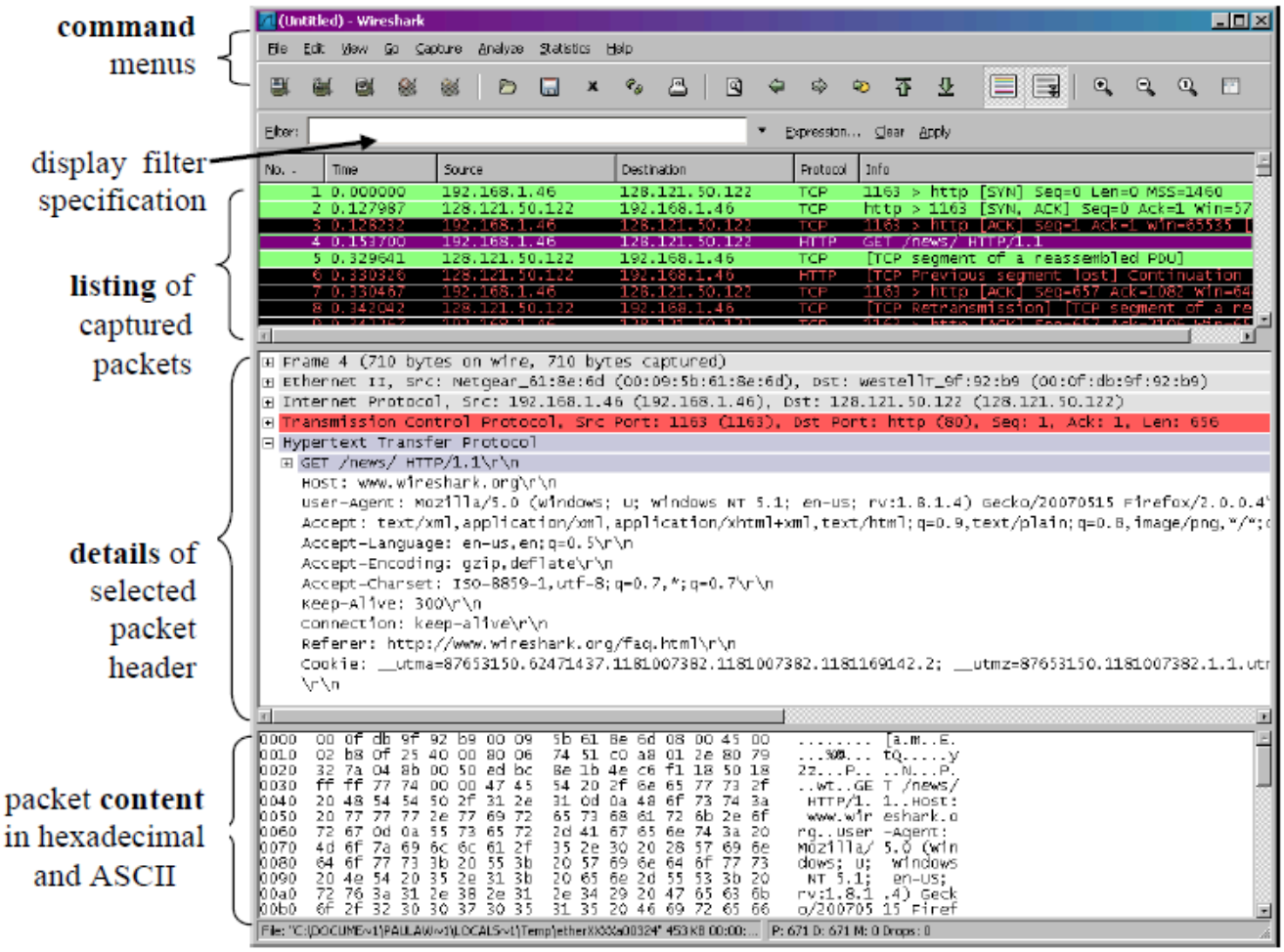


Figure: Wireshark Graphical User Interface

Taking Wireshark for a Test Run:

1. Open Wireshark, select interface from list of interfaces (Ethernet in your case). Then Press the capture Start button to start capturing the internet traffic packets at run time.
   If none of the interface is working you can use the Ethereal Packet Traces given in the Traces folder to answer the questions below (In this case you don't need to go through Steps 2 to 4 to capture the internet traffic as it is already done in the corresponding trace. To open the traffic in a trace go to File->Open and then select corresponding trace file)

2. While Wireshark is running, enter the URL: http://gaia.cs.umass.edu/wireshark-labs/INTRO-wireshark-file1.html and have that page displayed in your browser.

3. Now enter another URL http://gaia.cs.umass.edu/favicon.ico and you will see that this page is not found on the server.

4. In order to display both the pages, your browser will contact the HTTP server at gaia.cs.umass.edu and exchange HTTP messages with the server in order to download this page. The Ethernet frames containing these HTTP messages will be captured by Wireshark.

Note:

Make a Word file and post the screen shots of all the answers in it. Apart from the answers explore different settings of Wireshark and analyze all the layers of the Packets. Implement different filters in your data to view different grouping of packets. Make yourself familiar with the use of Wireshark.

Wireshark Filters:

- **Application Layer**: http, https
- **Transport Layer:** udp, udp.port==80, udp.srcport==443, udp.dstport==53
  tcp, udp.port==80, tcp.srcport==443, tcp.dstport==53
- **Network Layer:** ip.addr==192.168.1.2, ip.src==127.1.1.1, ip.dst==129.1.1.1
- **Data-Link Layer:** eth.addr == 00:00:5e:00:53:00, eth.src == 00:00:5e:00:53:00,
  eth.dst == 00:00:5e:00:53:00

To make combinations of different filters, you can use (&&, ||, !) signs as used in C++ e.g.,

a. http && tcp.port==80
b. tcp || ip.addr==192.168.1.2
c. !(ip.src==192.168.1.2)

In-Lab Statement 1: *Analyzing HTTP Protocol*

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The HTTP GET/Response Interaction

Use the http-ethereal-trace-1 packet trace to answer the questions below apply the "http" filter

a. You have packet data that contains all protocol messages exchanged between your computer and other network entities! The HTTP message exchanges with the gaia.cs.umass.edu web server should appear somewhere in the listing of packets captured. But there will be many other types of packets displayed as well.

b. Type in **"http"** (without the quotes, and in lower case – all protocol names are in lower case in Wireshark) into the display filter specification window at the top of the main Wireshark window. Then select Apply (to the right of where you entered "http"). **This will cause only HTTP message to be displayed in the packet-listing window.**

c. Select the first http message shown in the packet-listing window. This should be the **HTTP GET message that was sent from your computer to the gaia.cs.umass.edu HTTP server**. When you select the HTTP GET message, the Ethernet frame, IP datagram, TCP segment, and HTTP message header information will be displayed in the packet-header window3. By clicking plus and- minus boxes to the left side of the packet details window, minimize the amount of Frame, Ethernet, Internet Protocol, and Transmission Control Protocol information displayed. Maximize the amount information displayed about the HTTP protocol. Your Wireshark display should now look roughly as shown in the figure above.

d. Now try to find out the packet which contains the second request you sent to the browser and also analyze the packet which your browser received as a result of second GET Request and answer the following questions:

1- List up to **4 different protocols** that appear in the protocol column in the unfiltered packet-listing window.
2- How long did it take from when the HTTP GET message was sent until the HTTP OK reply was received?
3- Was the second Get Request successful? How can you tell it from the corresponding response packet?

By looking at the information in the HTTP GET and Response Messages for **BOTH the HTTP Requests**, answer the following questions

4. Is your **browser** running HTTP version 1.0 or 1.1? What **version** of HTTP is the server running?
5. What **languages** (if any) does your **browser** indicate that it can accept to the server?
6. What is the **IP address** of the gaia.cs.umass.edu server and your computer?
7. Identify the source and destination MAC addresses in the Ethernet frame of the HTTP GET request. Explain what these MAC addresses represent.
8. Identify the source port and destination port in the HTTP GET request. Which port is used by the HTTP server, and why?
9. What is the **status code** returned from the server to your browser?
10. When was the HTML file, that you are retrieving, **last modified** at the **server?**
11. What is the `Content-Length` of the HTTP response? Report the value in bytes.

The HTTP CONDITIONAL GET/response interaction

Use the http-ethereal-trace-2 packet trace to answer the questions below and apply the "http" filter

Answer the following questions:

12. Identify the `If-Modified-Since` header in the HTTP GET request. What date and time are specified in this header?
13. Inspect the HTTP response corresponding to this GET request. What is the HTTP status code and reason phrase returned by the server?
14. Examine the response packet and determine whether the requested object/content is returned by the server. Support your answer using the response status code and packet contents.
15. Compare the first and second GET requests in the trace. What is the difference between the two requests, and why does the server respond differently to the second request?

Post Lab

In-Lab Statement 2 : *Analyzing HTTP Protocol*

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Retrieving Long Documents

In our examples thus far, the documents retrieved have been simple and short HTML files. Let's next see what happens when we **download a long HTML file**. Do the following:

In the packet-listing window, you should see your HTTP GET message, followed by a **multiple-packet TCP response** to your HTTP GET request. This multiple-packet response deserves a bit of explanation. The **HTTP RESPONSE MESSAGE** consists of a status line, followed by header lines, followed by a blank line, followed by the entity body. In the case of our HTTP GET, the entity body in the response is the *entire* requested HTML file. In our case here, the HTML file is rather long, and at **4500 bytes is too large to fit in one TCP packet.** The single HTTP response message is thus broken into several pieces by TCP, with each piece being contained within a separate TCP segment. In recent versions of Wireshark, Wireshark indicates each **TCP segment as a separate packet**, and the fact that the single HTTP response was

fragmented across multiple TCP packets is indicated by the **"TCP segment of a reassembled PDU"** in the Info column of the Wireshark display.

- **Use the http-ethereal-trace-3 packet trace to answer the questions below and apply the "http" filter**

Answer the following questions:

1. How many HTTP GET request messages did your browser send?
2. Which **packet number** in the trace contains the GET message for **The Bill of Rights**?
3. Which **packet number** in the trace contains the status code and phrase associated with the response to the HTTP GET request?
4. What is the status code and phrase in the response?
5. How many data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights? What are the numbers of those packets?
6. The HTTP object has a total size of approximately 4500 bytes, but one of the packets shown in the trace contains only 490 bytes. Explain why a single packet does not contain the complete HTTP response. Use the TCP segment lengths in the trace to justify your answer.